WEEK -11

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 18-04-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 11)

B. **Sprint / Issue Goal:** To perform unit testing and integration testing, validate complete system workflow, and ensure all components function correctly.

C. **Status:** On Track

2. UML & OOAD Progress

A. **Diagrams Completed this week:** No new diagrams created

B. **Diagrams Revised:** No revisions

C. **OO Concept Applied:**

1. Modular testing of components
2. Validation of layered architecture (Frontend–Backend–Database–ML model)

3.Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

1. Performed backend method testing
2. Conducted integration testing across system components
3. Executed end-to-end system testing
4. Debugged and fixed issues identified during testing
5. Generated and analyzed test reports

Link://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

1. Updated test files and scripts
2. Committed bug fixes and improvements
3. Maintained version control for testing phase

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

C. **CI/CD Status:**

1. Testing performed manually.

4.Testing Summary

1. Performed **Unit Testing** to verify backend methods and functions
2. Conducted **Integration Testing** for:
   - Frontend → Backend communication
   - Backend → Database operations
   - Backend → ML model predictions
3. Validated **end-to-end system workflow**
4. Identified and resolved issues during testing
5. Generated and reviewed **test reports** for performance and correctness

5.Goals for Next Week

1. Finalize the **Deployment Diagram** to match the production environment
2. Prepare the application for deployment
3. Deploy the system to a **staging or production environment** (e.g., AWS / Azure / Heroku)
4. Verify application performance after deployment

Mentor Signature **Project Guide Signature**